Fault Classification and Localization in Distribution Networks Under Sparse Measurements Using Graph-Based Methods

Wei Sun, *Senior Member, IEEE*, Song Liu, Hengfeng Zhang, Zhiqiang Zhang, and Fei Lin

***Abstract*—Efficient fault classification and localization are crucial for rapid and reliable maintenance in distribution networks (DNs). However, conventional model-based approaches often lack robustness and generalizability due to the complex multibranch topology and time-varying operating conditions of DNs. Traditional machine-learning methods also underutilize the available data. In contrast, graph neural networks (GNNs) can capture richer, topology-aware fault features by aggregating information from neighboring nodes. Yet, sparse measurement deployment in DNs severely limits neighborhood information, and deeper GNNs often suffer from oversmoothing, which further degrades performance. To address these challenges, we propose a two-stage framework for fault classification and localization that integrates an extended feature propagation (EFP) algorithm and a residual graph convolutional network with neighbor filtering (RGCN-NF). First, EFP reconstructs features for unmeasurable nodes, alleviating the impact of sparse measurements and enriching local neighborhood information. Then, RGCN-NF incorporates neighbor filtering and node-level residual connections to support deeper architectures while mitigating oversmoothing. Extensive experiments on DN test cases across diverse operating scenarios, while accounting for practical measurement uncertainties, demonstrate that the proposed method consistently outperforms existing approaches and maintains high accuracy and strong robustness in fault classification and localization even under sparse measurements.**

***Index Terms*—Distribution networks (DNs), extended feature propagation (EFP), fault classification and localization, neighbor filtering, residual graph convolution network.**

I. Introduction

DISTRIBUTION networks (DNs) deliver electrical energy from transmission networks to end users, serving as a vital part of the power system [1], [2]. In recent years, the frequency of faults in DNs has increased due to aging infrastructure, greater integration of distributed energy resources, and more frequent extreme weather events [3], [4]. When fault clearance is delayed, it disrupts the local power supply and prolongs outages, leading to economic losses and safety risks [5], [6], [7]. Therefore, effective fault classification and localization techniques are essential for expediting repairs, shortening outage duration, and enhancing the safety and reliability of DNs.

Fault classification and localization methods for DNs can be broadly divided into model-based and machine-learning-based approaches. Among the model-based methods, Alwash et al. [8] proposed an improved impedance-based fault-location scheme for locating the exact fault point in unbalanced distribution systems with distributed generation. Mirshekali et al. [9] combined impedance calculations, distributed-parameter modeling, and least-squares matching for fault localization. Conte et al. [10] developed an impedance-estimation-based fault-localization algorithm with adjustable localization resolution. Chen et al. [11] employed a Hough-transform-enhanced single-ended traveling-wave method to improve wavefront identification accuracy and noise immunity. Wang et al. [12] used advanced signal-processing techniques to detect the initiation time of fault traveling waves, improving fault-location accuracy and reliability. Zhu and Pong [13] used magnetic induction detection of the decaying DC component in fault current for fault classification, while Dubey and Jena [14]. proposed a differential positive-sequence impedance-energy model for fault classification Despite these advances, model-based approaches often exhibit limited robustness under the complex, time-varying operating conditions of modern DNs.

In contrast, machine-learning-based methods have demonstrated strong performance in fault classification and localization in DNs, with robust generalization across a wide range of operating conditions and network configurations. Yang et al. [15] designed a fast multibranch residual network (Fa-Mb-ResNet) for fault identification and line selection, demonstrating its effectiveness under varying conditions. Convolutional neural networks (CNNs) have also been widely used for fault classification and localization [16], [17], [18]. Ghaemi et al. [19] used a random forest to classify and locate faults, validating the model's performance across multiple systems. Li et al. [20] employed Takagi–Sugeno fuzzy neural networks in combination with a genetic learning adaptive algorithm to diagnose faults in DNs. Furthermore, Li et al. [21]

Received 11 December 2025; revised 24 February 2026; accepted 27 February 2026. Date of publication 16 March 2026; date of current version 23 March 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 52277087, Grant 62173120, and Grant 62303149; and in part by Anhui Provincial Natural Science Foundation under Grant 2108085UD07 and Grant 2108085QE237. The Associate Editor coordinating the review process was Dr. Marco Carratù. *(Corresponding author: Zhiqiang Zhang.)*

Wei Sun is with Hefei University of Technology, Hefei 230009, China, and also with the State Key Laboratory of High-Efficiency and High-Quality Conversion for Electric Power, Hefei 230009, China (e-mail: wsun@hfut.edu.cn).

Song Liu, Hengfeng Zhang, and Zhiqiang Zhang are with Hefei University of Technology, Hefei 230009, China (e-mail: ls0313@mail.hfut.edu.cn; zhfz@mail.hfut.edu.cn; zhiqiangzhang@hfut.edu.cn).

Fei Lin is with the Marketing Service Center, State Grid Anhui Electric Power Company Ltd., Hefei 230000, China (e-mail: 4247227@qq.com).

Digital Object Identifier 10.1109/TIM.2026.3674271

1557-9662 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: XIDIAN UNIVERSITY. Downloaded on October 23,2026 at 05:39:18 UTC from IEEE Xplore. Restrictions apply.

embedded wavelet transform into a long short-term memory network to extract time–frequency features of fault signals to achieve high-accuracy early fault detection. However, these methods overlook the explicit topology of DNs, leading to inadequate feature extraction for fault classification and localization and preventing machine-learning models from achieving high-precision results.

Graph neural networks (GNNs), which explicitly model graph-structured data and internode relationships, have been applied to fault classification and localization in DNs and have achieved state-of-the-art performance [22]. Chen et al. [23] employed a graph convolutional network (GCN) for fault localization in DNs and verified its effectiveness from multiple perspectives. de Freitas and Coelho [24] proposed a gated GNN-based fault localization method that generalizes to network reconfiguration scenarios and identifies fault regions. Ngo et al. [25] utilized a 1-D convolutional graph attention network that leverages voltages and branch currents as features for fault detection and identification. Hu et al. [26] introduced a spatiotemporal graph learning model for fault classification and localization by incorporating specialized spatiotemporal convolutional blocks to extract both data-driven and structural features. Chanda and Soltani [27] applied a heterogeneous multitask GNN to jointly detect, locate, and classify faults, while also estimating fault resistance and current.

However, GNN-based methods still face two key limitations. First, many GNN-based methods [24], [25], [26] assume full instrumentation of all buses, which entails prohibitive equipment and communication costs [28], [29]. Conversely, when only a subset of buses is monitored [23], [27], information is restricted to those nodes, thus yielding incomplete feature representations. Second, studies [23], [24], [25], [26], [27] have adopted relatively shallow network architectures. However, this limits the models' expressive capacity, particularly when dealing with large-scale and complex graphs. Simply stacking more layers is not an effective solution; instead, it exacerbates the oversmoothing problem. Oversmoothing refers to the phenomenon where, as the number of GNN layers increases, nodes repeatedly aggregate information from their neighbors, causing their features to become increasingly similar and eventually indistinguishable, which significantly degrades overall performance [30], [31], [32].

To address both the constraint of sparse measurements and the challenge of oversmoothing in deep GNNs, we propose a two-stage framework that integrates an extended feature propagation (EFP) algorithm with a residual graph convolutional network with neighbor filtering (RGCN-NF). The proposed framework leverages voltage measurements from a subset of buses together with DN topology information to achieve high-precision fault classification and localization under sparse measurements. Through a neighbor-filtering mechanism and node-level residual connections, RGCN-NF preserves critical feature distinctions both within and across layers, ensuring robust performance under sparse measurement conditions. The main contributions can be summarized as follows.

1) To mitigate the impact of sparse measurements in DNs on fault classification and localization, we propose an EFP algorithm based on the FP algorithm, which reconstructs the features of unmeasurable nodes using information propagated from their measurable neighbors. Furthermore, to avoid degree-1 unmeasurable nodes (with only one neighbor) obtaining nearly identical features when relying solely on that neighbor, we enlarge their receptive fields to aggregate information from additional neighbors, thereby improving the effectiveness of data completion.
2) We propose RGCN-NF to mitigate oversmoothing and enable deeper feature extraction. This model filters out neighbors with highly dissimilar features before aggregation and introduces node-level residual connections across layers to stabilize feature updates. By preserving internode discriminability during deep propagation, RGCN-NF effectively alleviates oversmoothing.

The remainder of this article is organized as follows. Section II outlines the problem statement and introduces the graph structure of DNs. Section III presents the proposed fault classification and localization framework. Section IV validates the effectiveness and advantages of the proposed framework through experiments. Section V concludes this article.

II. Problem Formulation and Modeling

In this section, we first introduce the problem that this article aims to address. Then, we present the graph-based modeling of DNs.

A. Problem Statement

Various faults can occur in DNs, with power line faults being among the most common and impactful. Power line faults can be further classified into four main types: single-phase-to-ground faults, two-phase short-circuit faults, two-phase-to-ground faults, and three-phase-to-ground faults. Among them, single-phase-to-ground faults are the most prevalent, accounting for 70% of all power line faults in DNs. They are typically caused by insulation degradation, lightning strikes, or external damage, leading to a conductive path between the conductor and the ground. Two-phase short-circuit faults and two-phase-to-ground faults occur at rates of 15% and 10%, respectively. Three-phase-to-ground faults have the lowest occurrence rate, at 5%, but typically cause the most severe impact, potentially leading to widespread power outages [33]. The rms voltage profiles of these typical fault events are shown in Fig. 1. A deeper analysis of these fault characteristics is presented as follows.

1) Voltage abnormalities caused by faults vary depending on the fault type and location.
2) The extent to which faults affect different buses is influenced by the DN topology.

Therefore, the ultimate goal of our work is to determine the fault type and location by utilizing voltage variation characteristics and topological relationships during faults.

B. Graph Structure Modeling of DNs

A simple DN is shown in Fig. 2, consisting of buses, power lines, and loads. Using this DN as an example,

Authorized licensed use limited to: XIDIAN UNIVERSITY. Downloaded on October 23,2026 at 05:39:18 UTC from IEEE Xplore. Restrictions apply.

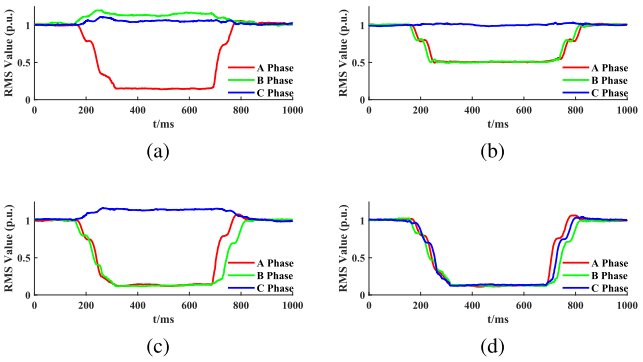


Fig. 1. RMS voltage profiles during typical fault events. (a) Single-phase-to-ground fault. (b) Two-phase short-circuit fault. (c) Two-phase-to-ground fault. (d) Three-phase-to-ground fault.

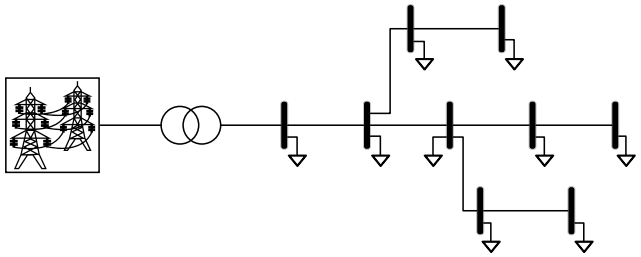


Fig. 2. Illustration of a simple DN.

we derive a general graph-theoretic model for DNs. The buses in the DN correspond to nodes in the graph, while the power lines between buses serve as edges connecting these nodes. Therefore, the DN can be represented as an undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{A})$, where the node set is $\mathcal{V} = \{v_1, v_2, \ldots, v_N\}$, and N denotes the number of nodes. The edge set is defined as $\mathcal{E} = \{e_{ij} | i, j \in N, \text{if a power line exists between } v_i \text{and } v_j\}$. The element a_{ij} of the adjacency matrix $\mathbf{A} \in \mathbb{R}^{N \times N}$ is defined as follows:

$$a_{ij} = \begin{cases} 1, & \text{if } e_{ij} \text{ exists} \\ 0, & \text{otherwise.} \end{cases} \tag{1}$$

In addition, the feature matrix of the nodes is denoted as $\mathbf{X} \in \mathbb{R}^{N \times K}$, where K represents the dimension of the node features. Since voltage magnitudes can only be measured on a subset of buses, the feature values corresponding to unmeasured nodes are set to zero in $\mathbf{X}$.

III. Proposed Fault Classification and Localization Framework

In this section, we introduce a two-stage fault classification and localization framework for DNs (see Fig. 3). First, an EFP algorithm reconstructs missing node measurements from available sensor data. Then, the RGCN-NF extracts diagnostic features and feeds them into two parallel fully connected branches for fault classification and localization, respectively.

A. EFP Algorithm

The FP algorithm reconstructs the features of unmeasurable nodes by iteratively solving a diffusion equation derived from the minimization of the Dirichlet energy, thereby enabling the effective application of downstream GNNs [34]. However, when the graph contains unmeasurable nodes with degree 1, FP can only utilize information from their single neighboring node. As the iteration converges, these nodes tend to have nearly identical features to their neighbors, which increases feature homogeneity and aggravates the oversmoothing problem in downstream GNNs, thus degrading model performance.

To address this limitation, we propose the EFP algorithm. Fig. 4 illustrates the feature propagation processes of FP and EFP. In the FP algorithm, the unmeasurable node 1 receives information only from its direct neighbor node 3. After several iterations, node 1's feature becomes nearly identical to that of node 3, leading to feature homogeneity and the oversmoothing issue, which reduces the discriminative capability and performance of downstream GNNs. In contrast, the EFP algorithm expands the information receptive field, allowing the degree-1 unmeasurable node 1 to receive information not only from its direct neighbor node 3 but also from multihop neighbors (nodes 2, 4, and 6). This extension allows node features to be represented more comprehensively, thereby alleviating the homogeneity effect and improving the performance of downstream tasks. The EFP algorithm is explained in detail below.

The adjacency matrix with self-loops is defined as the symmetric normalized adjacency matrix $\widetilde{\mathbf{A}} = \mathbf{D}^{-1/2}(\mathbf{A} + \mathbf{I})\mathbf{D}^{-1/2}$, where $\mathbf{D} = \text{diag}(d_1, d_2, \ldots, d_i, \ldots, d_N)$ and $d_i = \sum_{j=1}^{N} a_{ij} + 1$. The Laplacian matrix is given by $\boldsymbol{\Delta} = \mathbf{I} - \widetilde{\mathbf{A}}$. For convenience in this section, $\mathbf{X}$, $\widetilde{\mathbf{A}}$, and $\boldsymbol{\Delta}$ are rearranged as follows:

$$\mathbf{X} = \begin{bmatrix} \mathbf{X}_r \\ \mathbf{X}_s \\ \mathbf{X}_t \end{bmatrix}, \quad \widetilde{\mathbf{A}} = \begin{bmatrix} \widetilde{\mathbf{A}}_{rr} & \widetilde{\mathbf{A}}_{rs} & \widetilde{\mathbf{A}}_{rt} \\ \widetilde{\mathbf{A}}_{sr} & \widetilde{\mathbf{A}}_{ss} & \widetilde{\mathbf{A}}_{st} \\ \widetilde{\mathbf{A}}_{tr} & \widetilde{\mathbf{A}}_{ts} & \widetilde{\mathbf{A}}_{tt} \end{bmatrix}, \quad \boldsymbol{\Delta} = \begin{bmatrix} \boldsymbol{\Delta}_{rr} & \boldsymbol{\Delta}_{rs} & \boldsymbol{\Delta}_{rt} \\ \boldsymbol{\Delta}_{sr} & \boldsymbol{\Delta}_{ss} & \boldsymbol{\Delta}_{st} \\ \boldsymbol{\Delta}_{tr} & \boldsymbol{\Delta}_{ts} & \boldsymbol{\Delta}_{tt} \end{bmatrix} \tag{2}$$

where $\mathbf{X}_r$, $\mathbf{X}_s$, $\mathbf{X}_t$ represent the features of measurable nodes, unmeasurable nodes with a degree greater than 1, and degree-1 unmeasurable nodes, respectively.

Considering that $\widetilde{\mathbf{A}}$ determines the source of information for reconstructing features of unmeasurable nodes, when degree-1 nodes are unmeasurable, their feature reconstruction relies solely on a single neighbor. As the iterative computation converges, such nodes' features collapse toward those of their neighbor, exacerbating homogenization. To prevent $\mathbf{X}_t$ from matching exactly its sole neighbor's features, we expand the receptive field of degree-1 unmeasurable nodes to the kth order, where k is determined by the graph's complexity. The adjacency matrix is represented as follows:

$$\breve{\mathbf{A}} = \begin{bmatrix} \widetilde{\mathbf{A}}_{rr} & \widetilde{\mathbf{A}}_{rs} & \widetilde{\mathbf{A}}_{rt} \\ \widetilde{\mathbf{A}}_{sr} & \widetilde{\mathbf{A}}_{ss} & \widetilde{\mathbf{A}}_{st} \\ \widetilde{\mathbf{A}}^{k}_{tr} & \widetilde{\mathbf{A}}^{k}_{ts} & \widetilde{\mathbf{A}}^{k}_{tt} \end{bmatrix} \tag{3}$$

where $\widetilde{\mathbf{A}}^{k}_{tr}$ denotes the submatrix of the kth power of $\widetilde{\mathbf{A}}$ formed by the rows corresponding to $\mathbf{X}_t$ and the columns corresponding to $\mathbf{X}_r$. The Laplacian matrix is reformulated as $\breve{\boldsymbol{\Delta}} = \mathbf{I} - \breve{\mathbf{A}}$.

We assume that nodes in the graph adhere to a similarity principle, meaning that adjacent nodes have similar features. This similarity holds for bus voltage magnitudes in DNs, as the impedance of power lines causes voltage drops between buses,

Authorized licensed use limited to: XIDIAN UNIVERSITY. Downloaded on October 23,2026 at 05:39:18 UTC from IEEE Xplore. Restrictions apply.

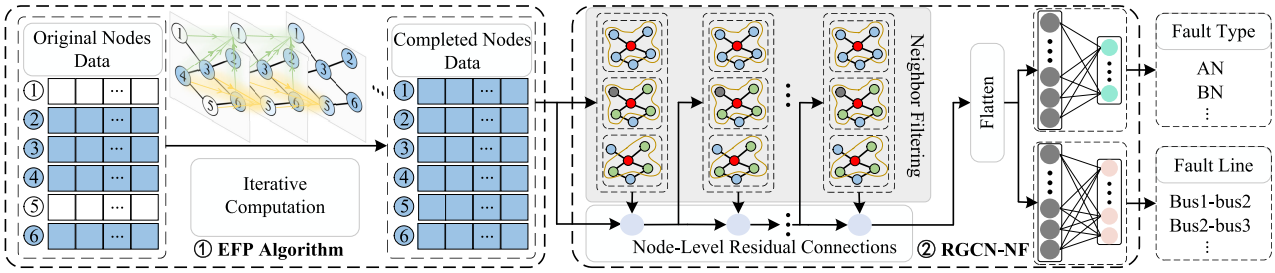


Fig. 3. Overall architecture of the proposed two-stage framework for fault classification and localization.

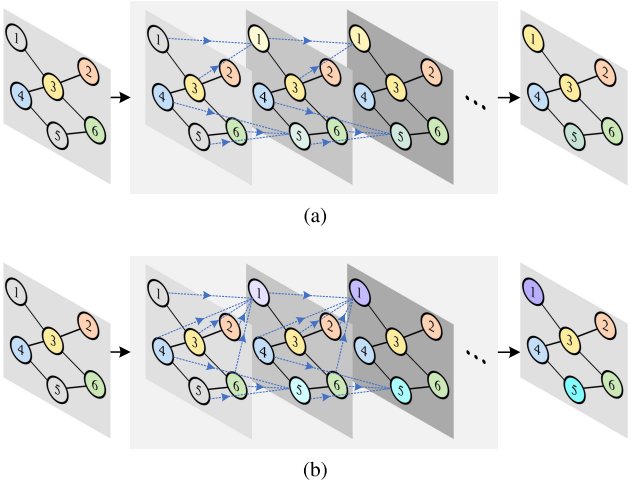


Fig. 4. Comparison of feature propagation in (a) FP and (b) EFP.

resulting in greater voltage similarity between adjacent buses. To quantify this similarity, Dirichlet energy is introduced. A reduction in Dirichlet energy promotes similarity between adjacent buses, thereby minimizing the Dirichlet energy to optimally reconstruct the unmeasurable node features. The Dirichlet energy can be defined as follows:

$$\xi(\mathbf{X}, \mathcal{G}) = \frac{1}{2}\mathbf{X}^{\top}\breve{\mathbf{\Delta}}\mathbf{X} = \frac{1}{2}\sum_{ij}\breve{a}_{ij}\left(\mathbf{x}_i - \mathbf{x}_j\right)^2 \tag{4}$$

where $\breve{a}_{ij}$ represents the element in the ith row and jth column of $\breve{\mathbf{A}}$, and $\mathbf{x}_i$, $\mathbf{x}_j$ are the features of nodes i and j, respectively.

Given the convexity of the Dirichlet energy, its minimum can be derived using the gradient flow [34]. Specifically, the gradient flow equation $\dot{\mathbf{X}}(t) = -\mathbf{\Delta}\mathbf{X}(t)$, subject to the condition $\mathbf{X}_r(t) = \mathbf{X}_r$, can be rewritten as follows:

$$\begin{aligned}\begin{bmatrix}\dot{\mathbf{X}}_r(t)\\ \dot{\mathbf{X}}_s(t)\\ \dot{\mathbf{X}}_t(t)\end{bmatrix} &= -\begin{bmatrix}\mathbf{0} & \mathbf{0} & \mathbf{0}\\ \breve{\mathbf{\Delta}}_{sr} & \breve{\mathbf{\Delta}}_{ss} & \breve{\mathbf{\Delta}}_{st}\\ \breve{\mathbf{\Delta}}_{tr} & \breve{\mathbf{\Delta}}_{ts} & \breve{\mathbf{\Delta}}_{tt}\end{bmatrix}\begin{bmatrix}\mathbf{X}_r\\ \mathbf{X}_s(t)\\ \mathbf{X}_t(t)\end{bmatrix}\\ &= -\begin{bmatrix}\mathbf{0}\\ \breve{\mathbf{\Delta}}_{sr}\mathbf{X}_r + \breve{\mathbf{\Delta}}_{ss}\mathbf{X}_s(t) + \breve{\mathbf{\Delta}}_{st}\mathbf{X}_t(t)\\ \breve{\mathbf{\Delta}}_{tr}\mathbf{X}_r + \breve{\mathbf{\Delta}}_{ts}\mathbf{X}_s(t) + \breve{\mathbf{\Delta}}_{tt}\mathbf{X}_t(t)\end{bmatrix}.\end{aligned} \tag{5}$$

When $\dot{\mathbf{X}}(t) = \mathbf{0}$, the Dirichlet energy attains its minimum, yielding the optimal solution $\mathbf{X}$. However, solving this equation involves matrix inversion, which is computationally expensive. To address this, the equation is discretized, allowing $\mathbf{X}$ to be computed through iterative methods. By approximating the time derivative with forward differences and setting the step size to 1, we obtain the diffusion equation

$$\begin{aligned}\mathbf{X}^{(k+1)} &= \begin{bmatrix}\mathbf{I} & \mathbf{0} & \mathbf{0}\\ -\breve{\mathbf{\Delta}}_{sr} & \mathbf{I} - \breve{\mathbf{\Delta}}_{ss} & -\breve{\mathbf{\Delta}}_{st}\\ -\breve{\mathbf{\Delta}}_{tr} & -\breve{\mathbf{\Delta}}_{ts} & \mathbf{I} - \breve{\mathbf{\Delta}}_{tt}\end{bmatrix}\mathbf{X}^{(k)}\\ &= \begin{bmatrix}\mathbf{I} & \mathbf{0} & \mathbf{0}\\ \tilde{\mathbf{A}}_{sr} & \tilde{\mathbf{A}}_{ss} & \tilde{\mathbf{A}}_{st}\\ \tilde{\mathbf{A}}^{k}_{tr} & \tilde{\mathbf{A}}^{k}_{ts} & \tilde{\mathbf{A}}^{k}_{tt}\end{bmatrix}\mathbf{X}^{(k)}.\end{aligned} \tag{6}$$

According to (6), the unmeasurable node features can be reconstructed from the measurable ones.

B. Residual GCN With Neighbor Filtering

The first-order Chebyshev polynomial approximation of GCN effectively simplifies the computational complexity of spectral graph convolution and is widely used in graph classification tasks [35]. However, the oversmoothing problem in this model limits its performance. To mitigate oversmoothing, we propose the RGCN-NF algorithm. The RGCN-NF layer workflow is illustrated in Fig. 5 and detailed below.

Not all neighboring nodes contribute positively during aggregation, especially under fault conditions in DNs. Physically, buses that are electrically close to the fault and lie on or near the main fault-propagation path tend to exhibit similar voltage-sag patterns, whereas electrically distant or weakly coupled buses show much weaker or inconsistent responses. When a node's neighbors exhibit significant differences in their voltage-sag features, aggregating them mixes physically irrelevant information, causes substantial changes in the node's representation, and aggravates the oversmoothing problem. We therefore regard such neighbors as "harmful," since their aggregation obscures the physically meaningful fault-propagation pattern. To mitigate this effect and better align the learned representations with the underlying physical mechanism, we introduce a quantitative criterion to filter out harmful neighbors before aggregation. Specifically, the feature similarity $c_{u,v}^{(l+1)}$ between a node u and its neighbor v (including u itself) at layer $l+1$ is computed as follows:

$$c_{u,v}^{(l+1)} = \exp\left(-\left\|\mathbf{z}_u^{(l)} - \mathbf{z}_v^{(l)}\right\|_1\right) \tag{7}$$

where $\mathbf{z}_u^{(l)}$ and $\mathbf{z}_v^{(l)}$ denote the output features of nodes u and v at layer l. This similarity measures the consistency between their features. Similarity values below the threshold T_h are set to zero, thereby effectively filtering out "harmful" nodes

Authorized licensed use limited to: XIDIAN UNIVERSITY. Downloaded on October 23,2026 at 05:39:18 UTC from IEEE Xplore. Restrictions apply.

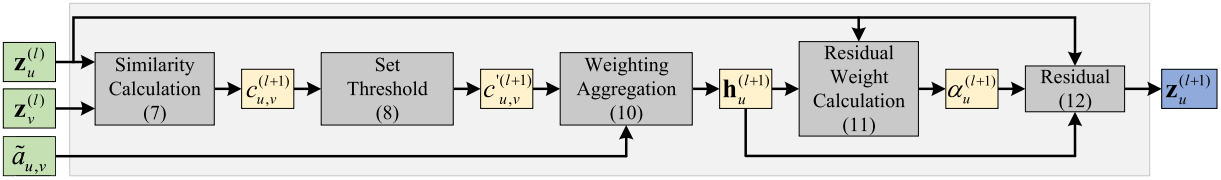


Fig. 5. RGCN-NF layer workflow.

during aggregation, as follows:

$$c_{u,v}^{\prime(l+1)} = \begin{cases} c_{u,v}^{(l+1)}, & \text{if } c_{u,v}^{(l+1)} > T_h \\ 0, & \text{otherwise} \end{cases} \tag{8}$$

where $c_{u,v}^{\prime(l+1)}$ represents the similarity score, which also serves as a weighting factor in the aggregation process. Nodes with higher similarity scores are assigned greater weights, thereby placing greater emphasis on features from similar neighbors during information propagation. Thus, the weighted adjacency matrix element $\tilde{a}_{u,v}^{\prime(l+1)}$ can be represented as follows:

$$\tilde{a}_{u,v}^{\prime(l+1)} = \frac{c_{u,v}^{\prime(l+1)} \tilde{a}_{u,v}}{\sum_{n=1}^{N} c_{u,n}^{\prime(l+1)} \tilde{a}_{u,n}}. \tag{9}$$

After assigning weights to the neighboring nodes, the aggregation process for node u is represented by

$$\mathbf{h}_u^{(l+1)} = \delta\left(\sum_{v \in N(u)} \tilde{a}_{u,v}^{\prime\,(l+1)} \mathbf{z}_v^{(l)} \mathbf{W}^{(l+1)} \right) \tag{10}$$

where $\mathbf{h}_u^{(l+1)}$ denotes the aggregated feature of node u at layer $l+1$. $\delta(\cdot)$ is an activation function. $N(u)$ represents the set of neighboring nodes of u, and $\mathbf{W}^{(l+1)}$ is the learnable parameter matrix of layer $l+1$.

Subsequently, node-level residual connections are introduced after aggregation. In this mechanism, the difference between the aggregated and input features of each node is used to measure the degree of feature variation, based on which individualized residual weights are assigned. Unlike conventional residual connections, node-level residual connections help preserve personalized node information. The residual weight $\alpha_u^{(l+1)}$ of node u at layer $l+1$ is defined as follows:

$$\alpha_u^{(l+1)} = \exp\left(- \left\| \mathbf{h}_u^{(l+1)} - \mathbf{z}_u^{(l)} \right\|_1 \right). \tag{11}$$

The residual weight is then used to update the output feature $\mathbf{z}_u^{(l+1)}$ of node u, as follows:

$$\mathbf{z}_u^{(l+1)} = \left(1 - \alpha_u^{(l+1)}\right) \mathbf{z}_u^{(l)} + \alpha_u^{(l+1)} \mathbf{h}_u^{(l+1)}. \tag{12}$$

When $\mathbf{h}_u^{(l+1)}$ and $\mathbf{z}_u^{(l)}$ are similar, $\alpha_u^{(l+1)}$ approaches 1, indicating that most features in $\mathbf{z}_u^{(l+1)}$ are derived from $\mathbf{h}_u^{(l+1)}$. Conversely, if there is a significant difference, $\alpha_u^{(l+1)}$ tends toward a lower value, retaining more features from $\mathbf{z}_u^{(l)}$.

We stack L RGCN-NF layers and assume the output of the Lth layer as $\mathbf{H}^{(L)}$. We then flatten $\mathbf{H}^{(L)}$ into a 1-D vector $\mathbf{p}$, and use two independent, parallel fully connected layers as the classifier for fault classification and localization, as follows:

$$\hat{\mathbf{y}}^1 = \text{softmax}\left(\mathbf{W}_f^1 \mathbf{p} + \mathbf{b}_1 \right) \tag{13}$$

$$\hat{\mathbf{y}}^2 = \text{softmax}\left(\mathbf{W}_f^2 \mathbf{p} + \mathbf{b}_2 \right) \tag{14}$$

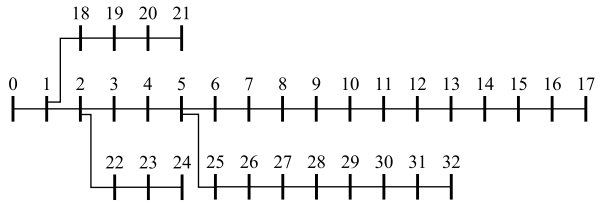


Fig. 6. Topology of the IEEE 33-bus system.

where $\hat{\mathbf{y}}^1$ and $\hat{\mathbf{y}}^2$ represent the predicted outputs for the fault classification and localization tasks, respectively. The parameters $\mathbf{W}_f^1$, $\mathbf{b}_1$, $\mathbf{W}_f^2$, and $\mathbf{b}_2$ are the weight matrices and biases associated with each task. This setup allows the model to handle both fault classification and localization simultaneously while meeting the dimensional requirements of each task. The loss is computed by applying weights to the cross-entropy losses for both tasks

$$L_1 = -\sum_{m=1}^{M} \mathbf{y}_m^1 \log\left(\hat{\mathbf{y}}_m^1 \right) \tag{15}$$

$$L_2 = -\sum_{q=1}^{Q} \mathbf{y}_q^2 \log\left(\hat{\mathbf{y}}_q^2 \right) \tag{16}$$

$$L_{\text{total}} = \beta L_1 + \gamma L_2 \tag{17}$$

where β and γ are the loss weights for the fault classification and localization tasks, while L_1 and L_2 represent the respective loss values for each task. M and Q denote the number of classes for each task. For fault classification, $\mathbf{y}_m^1$ and $\hat{\mathbf{y}}_m^1$ correspond to the true and predicted probabilities of the mth class. Similarly, $\mathbf{y}_q^2$ and $\hat{\mathbf{y}}_q^2$ represent the true and predicted probabilities of the qth class in the fault localization task.

IV. Case Study

This section experimentally evaluates the proposed fault classification and localization framework for DNs, with a focus on its accuracy under sparse measurements, robustness across different fault scenarios, and scalability across different test systems, all while accounting for practical measurement uncertainties.

A. Implementation Details

To evaluate the effectiveness of the proposed RGCN-NF fault classification and localization method based on the EFP algorithm, the IEEE 33-bus system [36] is modeled in PSCAD/EMTDC and used for experimental validation. The IEEE 33-bus system is shown in Fig. 6. It represents a typical

Authorized licensed use limited to: XIDIAN UNIVERSITY. Downloaded on October 23,2026 at 05:39:18 UTC from IEEE Xplore. Restrictions apply.

TABLE I
MEASUREMENT ERROR SETTINGS ACROSS OPERATING STAGES

Operating stage	VT measurement error		PMU measurement error		Communication loss (%)
	$\lvert\cdot\rvert$ (%)	$\angle$ (crad)	$\lvert\cdot\rvert$ (%)	$\angle$ (crad)	
Pre-fault steady state	3	3.5	0.1	0.1	5
During the fault	3	3.5	TVE3	TVE3	5
Post-fault transient	3	3.5	TVE3	TVE3	5
Post-fault steady state	3	3.5	0.1	0.1	5

TABLE II
BUS MEASUREMENT DEVICE DEPLOYMENT UNDER DIFFERENT DENSITIES

Device density	Buses with the measurement devices
70%	0, 1, 4, 5, 7, 8, 10, 11, 14, 15, 16, 17, 18, 20, 21, 22, 25, 26, 27, 28, 29, 31, 32
50%	1, 4, 5, 8, 10, 11, 12, 15, 16, 17, 18, 20, 24, 25, 26, 28, 31
30%	0, 2, 6, 10, 12, 16, 22, 23, 26, 28

medium-voltage radial DN, consisting of buses, power lines, and loads. The fault classification task, which includes four fault types, is formulated as a 4-class graph classification problem. Likewise, because the test system contains 32 power lines, fault localization is formulated as a 32-class graph classification problem.

To account for different measurement uncertainties and operating conditions, the following assumptions are made for the experimental setup, with reference to [37], [38], [39], and [40].

1) Line parameters are based on their nominal values, with a maximum allowable deviation of $\pm 15\%$. In the experiment, the actual values of line parameters are randomly drawn from a uniform distribution centered at the nominal value with a deviation range of $\pm 15\%$.
2) The practical measurement chain includes voltage transformers (VTs) rated as accuracy class 3P, with a maximum ratio error of 3% and a maximum phase-angle error of 3.5 crad. These VT error settings are applied consistently across all operating stages.
3) The phasor measurement units (PMUs) provide 100 measurements per second. Under steady-state conditions (prefault steady state and postfault steady state), the PMUs have a maximum magnitude error of 0.1% and a maximum phase-angle error of 0.1 crad. Under dynamic conditions, i.e., during the fault-on interval and the post-fault dynamic recovery stage, the PMU measurement error is constrained by a maximum total vector error (TVE) of 3%.
4) Communication losses are also considered. To account for packet loss during data transmission, it is assumed that up to 5% of the samples from each PMU may be missing. The measurement-chain error settings under different operating conditions are summarized in Table I.
5) Under normal operating conditions, the system active power and reactive power fluctuate within $\pm 30\%$ of their respective nominal values.
6) Three measurement-device density levels are considered: 30%, 50%, and 70%. The device density is defined as the ratio of the number of buses equipped with measurement devices to the total number of buses in the DN. For each density level, the bus locations where the measurement devices are installed are listed in Table II.
7) The fault resistance R_f is set to 0.1–100 Ω. In the experimental scenarios, R_f follows a uniform distribution over this interval.
8) The fault type is randomly selected from single-phase-to-ground, two-phase short-circuit, two-phase-to-ground, and three-phase-to-ground faults, while the fault location is randomly assigned along the distribution lines.

Monte Carlo (MC) simulations are performed with $N_{MC} = 1000$. Each trial corresponds to a random realization of uncertainties, in which line-parameter deviations, VT and PMU errors, and communication losses are sampled from uniform distributions bounded by their respective maximum error limits. On this basis, each trial further generates $P = 50$ different test cases. In each test case, the measurement-device density, normal-operating load fluctuations, fault type, fault location, fault resistance R_f, fault occurrence time, and fault duration are randomly sampled; the fault duration is required to be no shorter than the minimum time resolution corresponding to the measurement reporting rate.

Based on the above, three evaluation metrics are adopted: fault classification accuracy (FCA), fault localization accuracy (FLA), and joint accuracy (JA)

$$\mathrm{FCA} = \frac{1}{N}\sum_{n=1}^{N}\mathbb{I}\left(\hat{y}_{cls}^{(n)} = y_{cls}^{(n)}\right) \times 100\% \tag{18}$$

$$\mathrm{FLA} = \frac{1}{N}\sum_{n=1}^{N}\mathbb{I}\left(\hat{y}_{line}^{(n)} = y_{line}^{(n)}\right) \times 100\% \tag{19}$$

$$\mathrm{JA} = \frac{1}{N}\sum_{n=1}^{N}\mathbb{I}\left(\hat{y}_{cls}^{(n)} = y_{cls}^{(n)} \wedge \hat{y}_{line}^{(n)} = y_{line}^{(n)}\right) \times 100\% \tag{20}$$

where N denotes the total number of evaluated cases. For the nth case, $y_{cls}^{(n)}$ is the ground-truth fault type label and $\hat{y}_{cls}^{(n)}$ is the predicted fault type; $y_{line}^{(n)}$ is the ground-truth faulted-line label and $\hat{y}_{line}^{(n)}$ is the predicted faulted line. $\mathbb{I}(\cdot)$ is the indicator function, which equals 1 if the condition in the parentheses holds, and 0 otherwise.

The dataset is split into 60% for training, 20% for validation, and 20% for testing. Based on theoretical and experimental analyses, supported by prior literature and a grid search, the parameters of EFP and RGCN-NF are finalized as follows.

Authorized licensed use limited to: XIDIAN UNIVERSITY. Downloaded on October 23,2026 at 05:39:18 UTC from IEEE Xplore. Restrictions apply.

TABLE III
TEST-SET ACCURACY AND INFERENCE TIME OF DIFFERENT MODELS ON THE IEEE 33-BUS SYSTEM

Models	FCA	FLA	JA	Time (s)
RGCN-NF	**94.42**	**93.17**	**90.51**	2.16×10^{-4}
STGCN [26]	91.67	86.59	83.23	3.34×10^{-4}
GCN [23]	89.61	81.96	76.49	$\mathbf{2.26\times10^{-5}}$
Fa-Mb-ResNet [15]	89.87	82.61	78.32	5.97×10^{-5}
CNN [18]	85.28	79.82	73.83	8.14×10^{-4}

1) For both the EFP algorithm and the RGCN-NF model, the stopping criteria are determined automatically rather than manually preset. The EFP algorithm monitors the norm of the difference between consecutive iterations and stops once this difference no longer decreases, indicating the natural convergence of its diffusion-based updates. Similarly, the RGCN-NF model adopts an early-stopping strategy based on validation performance, halting training when further epochs fail to improve generalization and begin to show signs of overfitting—that is, when the model fits the training data too closely and consequently generalizes poorly to unseen data.
2) The learning rate is set to 0.005 and decays by a factor of 0.95 every 10 epochs. This follows the fast-then-stable optimization principle [41], where a relatively large initial learning rate enables rapid early convergence, while gradual decay reduces oscillations and improves generalization.
3) The batch size and dropout rate are set to 64 and 0.6, respectively. These choices are supported by findings in [42] and [43] and further validated through empirical evaluation. A dropout rate of 0.6 reduces the model's reliance on specific neurons, thereby mitigating overfitting and improving generalization.
4) The optimal configuration uses $k = 3$, $T_h = 0.15$, $\beta = 0.3$, $\gamma = 0.7$, and a six-layer architecture. To determine these parameters, the depth is first fixed at two layers—consistent with conventional GCN design [35]—and a grid search is performed over k, T_h, β, and γ under the constraint $\beta + \gamma = 1$ to ensure stable loss scaling. After the optimal values are identified, the impact of depth is evaluated: performance improves as the number of layers increases from 2 to 6, and then gradually declines when it exceeds 6 layers.

All experiments are conducted on the following hardware configuration: CPU: AMD Ryzen 7 6800H, GPU: NVIDIA GeForce RTX 3060, and Memory: 16 GB RAM.

B. Verifying the Effectiveness of RGCN-NF

To evaluate the effectiveness of RGCN-NF, it is compared with several high-performing baseline models for fault classification and localization in DNs, including STGCN [26], GCN [23], Fa-Mb-ResNet [15], and CNN [18]. Notably, all models take as input the data completed by the EFP algorithm. Table III reports the FCA, FLA, JA, and the inference execution time of each model on the test set. Here, the inference execution time is defined as the time required, after training, to complete a single forward pass for one test sample and output the final decision. Specifically, for each test sample, the elapsed time is measured from the moment the sample features are fed into the network to the moment the predicted result is produced, and the average inference time over all test samples is reported in the table.

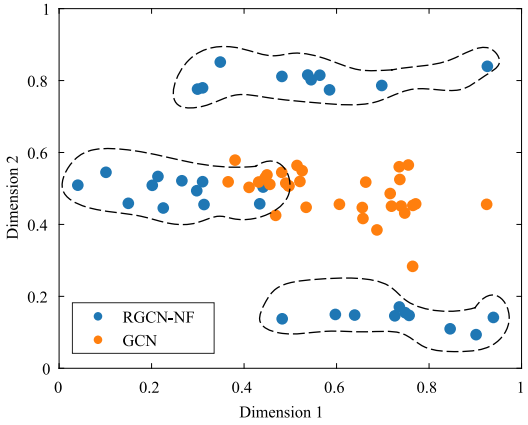


Fig. 7. t-SNE visualization of output features from RGCN-NF and GCN.

Among the five models compared in Table III, the proposed RGCN-NF achieves the highest accuracy for both fault classification and fault localization, as well as for the joint task, demonstrating its overall effectiveness. Notably, because fault localization is more challenging than fault classification, the other four models exhibit a pronounced performance gap between these two tasks. In contrast, RGCN-NF shows a smaller gap, indicating more stable performance and a stronger capability in handling the more complex localization problem.

To further illustrate the effectiveness of RGCN-NF, the aggregated output features are analyzed. A single fault sample is randomly selected and separately fed into the RGCN-NF and GCN models. Fig. 7 presents the t-SNE visualization of the resulting feature representations after six aggregation layers. As shown, the GCN features tend to collapse into a more uniform distribution, which increases sample ambiguity. In contrast, RGCN-NF preserves clearer separations among certain nodes even after six layers, which is beneficial for graph classification. These observations indicate that neighbor filtering and node-level residual connections effectively alleviate the over smoothing issue.

C. Effectiveness Analysis of the EFP Algorithm

To evaluate the effectiveness of the EFP algorithm in completing data for unmeasurable nodes, experiments are conducted using RGCN-NF with three types of inputs: data completed by the EFP algorithm, data completed by the FP algorithm, and incomplete data. Fig. 8 presents the accuracy of the test set. Under various measurement device densities, the EFP algorithm ensures that RGCN-NF maintains high accuracy in fault classification and localization. However, as the device density decreases, the accuracy gap between the FP and EFP algorithms increases. This is due to the increased number

Authorized licensed use limited to: XIDIAN UNIVERSITY. Downloaded on October 23,2026 at 05:39:18 UTC from IEEE Xplore. Restrictions apply.

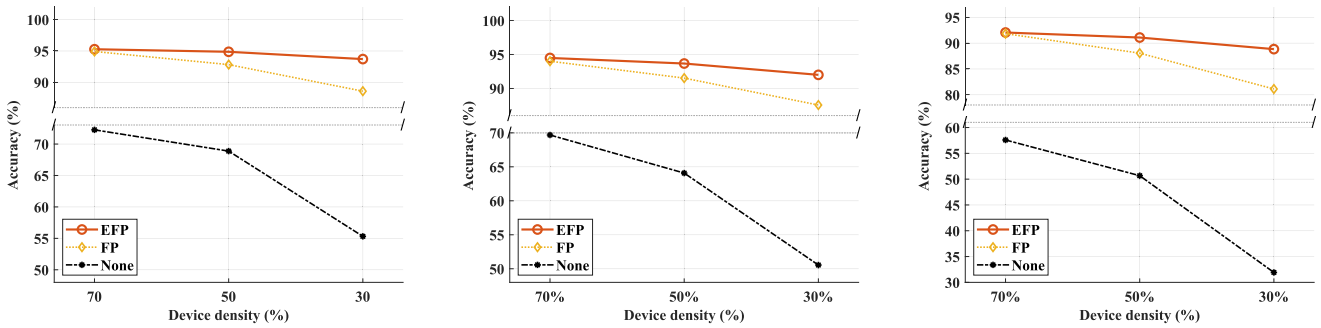


Fig. 8. Performance comparison of data completion methods under different measurement device densities on the IEEE 33-bus system. (a) FCA. (b) FLA. (c) JA.

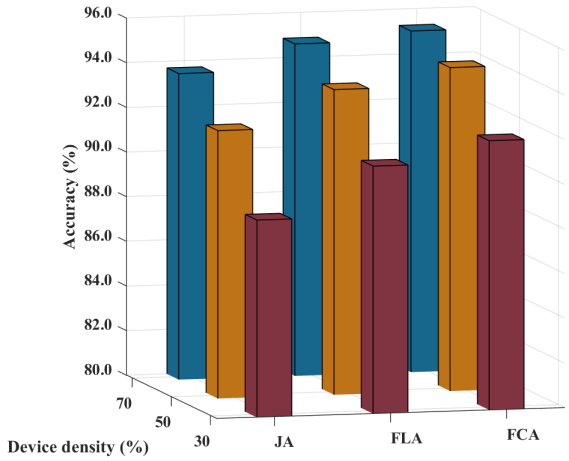


Fig. 9. Test accuracy of EFP under random measurement-device placements.

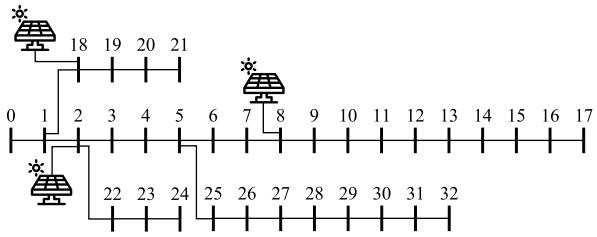


Fig. 10. IEEE 33-bus system with integrated PV units.

TABLE IV
TEST-SET ACCURACY OF DIFFERENT MODELS ON THE IEEE 33-BUS SYSTEM WITH PV INTEGRATION

Models	FCA	FLA	JA
RGCN-NF	**93.19**	**91.43**	**88.71**
STGCN [26]	88.79	83.20	78.28
GCN [23]	85.63	74.62	69.70
Fa-Mb-ResNet [15]	87.15	75.86	71.39
CNN [18]	80.58	76.17	65.85

of degree-1 unmeasurable nodes, which limits the performance of the FP algorithm. Moreover, without data completion, both FCA and FLA drop significantly because measurable nodes cannot aggregate information from unmeasurable neighbors, thereby limiting the performance of RGCN-NF. These results confirm that the EFP algorithm is effective in mitigating the accuracy loss in fault classification and localization caused by sparse measurements.

In addition, to evaluate the generalization capability of the proposed EFP algorithm under varying measurement-device placements, an additional set of 1000 test samples is generated for each measurement-device density level following the same data-generation procedure described above. The key difference is that, for each test case, the measurement-device locations are randomly resampled under the corresponding density constraint to emulate practical scenarios in which device deployment may be uncertain or adjusted. As shown in Fig. 9, both the single-task and joint-task test-set accuracies remain high after EFP-based completion, indicating that the EFP algorithm is robust to variations in measurement-device placement and generalizes well under such perturbations.

D. Performance Analysis of the Method With Renewable Energy Access

With the increasing integration of renewable energy units into DNs, new challenges have emerged for fault classification and localization. To emulate this scenario, photovoltaic (PV) units are incorporated into the IEEE 33-bus system for analysis. As shown in Fig. 10, PV units are connected at buses 2, 8, and 18, while the rest of the system remains unchanged. A PV penetration level of 30% is adopted, and the total PV capacity is equally allocated among the three PV units; thus, each unit accounts for one-third of the total PV capacity. Since variations in solar irradiance and module temperature typically evolve on a much slower time scale than the fault dynamics of the DN and the electromagnetic transient response of the inverter, their impacts are neglected in this study. Accordingly, the active-power reference of each PV unit is fixed at its rated output corresponding to the specified penetration level, ensuring consistency and reproducibility of the PV operating condition.

Subsequently, fault samples with integrated PV units are generated using the same MC simulation procedure and the same system configuration and parameter settings as described above. To evaluate the performance of the proposed model, comparisons are conducted with the baseline models. The FCA, FLA, and JA results are summarized in Table IV.

Authorized licensed use limited to: XIDIAN UNIVERSITY. Downloaded on October 23,2026 at 05:39:18 UTC from IEEE Xplore. Restrictions apply.

TABLE V
INDEPENDENT EVALUATION UNDER INDIVIDUAL MEASUREMENT UNCERTAINTY SOURCES

Models	Line-parameter deviation			VT measurement error			PMU measurement error			Communication loss			Ideal condition		
	FCA	FLA	JA	FCA	FLA	JA	FCA	FLA	JA	FCA	FLA	JA	FCA	FLA	JA
RGCN-NF	**99.53**	**97.71**	**97.50**	96.42	**94.63**	**93.15**	94.63	**93.87**	**91.05**	**99.65**	**97.19**	**97.03**	**99.96**	**98.73**	**98.70**
STGCN [26]	95.81	93.09	91.96	94.14	91.73	89.29	**95.06**	91.81	89.85	95.81	93.59	92.30	96.53	93.69	92.85
GCN [23]	93.96	92.17	87.52	92.81	86.53	84.71	90.26	86.29	80.87	93.17	91.93	87.79	95.92	92.20	89.76
Fa-Mb-ResNet [15]	93.21	86.17	83.36	**96.79**	88.46	87.07	91.94	86.45	81.19	98.42	87.85	86.90	98.61	90.36	90.17
CNN [18]	87.92	83.08	78.65	91.43	83.67	79.50	89.41	82.88	77.64	93.51	88.42	85.68	93.56	88.90	85.93

Although the integration of PV units increases the complexity of fault features, the proposed model still maintains higher accuracy than other models, demonstrating its effectiveness in DNs with renewable energy integration.

E. Sensitivity Analysis Under Individual Uncertainty Sources

To analyze the impact of different uncertainty sources on model performance, an independent evaluation is conducted using a single-uncertainty injection strategy. In each setting, only one source of uncertainty—line-parameter deviation, VT error, PMU error, or communication loss—is introduced, while all other sources are set to ideal conditions. The corresponding results are summarized in Table V. In addition, an ideal baseline is provided in which none of the four uncertainty sources is considered. It should be noted that this ideal case does not represent practical measurement environments; rather, it is used to verify that the proposed model can correctly perform fault classification and localization in the absence of measurement errors and to serve as an upper-bound reference. This facilitates quantifying the performance degradation induced by measurement uncertainties and provides a unified benchmark for comparison across different uncertainty configurations.

It can be observed that the proposed RGCN-NF outperforms the baseline models in most scenarios. Under line-parameter deviation, the GNN-based models (RGCN-NF, STGCN, and GCN) exhibit stronger robustness than the other models (Fa-Mb-ResNet and CNN). This is mainly because GNNs place greater emphasis on capturing the relative relationships among nodes. Although line-parameter deviations may cause an overall shift in voltage magnitudes and phase angles, they do not change the topological connectivity among nodes; therefore, their impact on GNNs is relatively limited. In addition, under communication-loss conditions, all five models show only slight performance degradation, largely benefiting from the imputation capability of the EFP algorithm: it not only recovers features of unmeasured nodes but also effectively reconstructs missing data points, thereby mitigating the impact of packet loss on model performance.

F. Robustness Evaluation Under High-Resistance Faults

High-resistance faults are typically characterized by weak fault signatures and low fault currents, and their measurement signals can be easily overwhelmed by noise, making them difficult to detect in DNs [44], [45]. To further evaluate the generalization capability of the proposed model under high-resistance conditions, the fault-resistance range is extended to 0.1–2000 Ω, and test samples are generated by uniformly sampling the fault resistance over this interval. The test results of the proposed model and the baseline models under high-resistance conditions are summarized in Table VI.

TABLE VI
MODEL PERFORMANCE UNDER HIGH-RESISTANCE FAULT CONDITIONS

Models	FCA	FLA	JA
RGCN-NF	**90.54**	**88.21**	**83.24**
STGCN [26]	85.25	79.60	72.31
GCN [23]	80.98	74.02	65.48
Fa-Mb-ResNet [15]	86.79	77.56	71.72
CNN [18]	76.90	70.12	59.49

Under high-resistance fault conditions, the performance of all models declines to varying degrees. As the fault resistance increases, the magnitude of node-voltage disturbances becomes smaller, reducing the amount of effective features available for learning and thereby further increasing the difficulty of fault classification and localization. Nevertheless, RGCN-NF achieves the best overall performance. The results indicate that, although high-resistance faults weaken the distinguishable characteristics of measurement signals, RGCN-NF can still effectively suppress noise and enhance internode feature differentiation through its neighbor-filtering mechanism and node-level residual connections, thus exhibiting strong robustness to high-resistance faults.

However, in practical DNs, the fault resistance is often unknown and may differ significantly from the range covered during model training [26]. To verify the model's generalization capability under a shifted resistance distribution, a cross-resistance-domain experiment is conducted: RGCN-NF is trained and validated using only low-resistance fault data (0.1–100 Ω), whereas the test set spans a much wider resistance range (0.1–2000 Ω). Specifically, the test set is partitioned into four resistance intervals (in Ω): (0.1, 500],

Authorized licensed use limited to: XIDIAN UNIVERSITY. Downloaded on October 23,2026 at 05:39:18 UTC from IEEE Xplore. Restrictions apply.

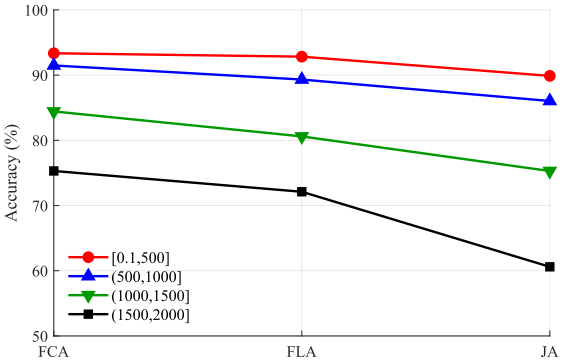


Fig. 11. Cross-resistance-domain test accuracy of RGCN-NF under different fault-resistance ranges.

(500, 1000], (1000, 1500], and (1500, 2000], and fault classification and localization performance is systematically evaluated across these resistance levels. The test accuracy under different fault-resistance ranges is shown in Fig. 11.

As the fault resistance increases, performance degradation becomes more pronounced. Within the (0.1, 500] Ω interval, the proposed model maintains relatively high FCA, FLA, and JA. As the resistance further increases, fault-induced measurable signal features gradually weaken, reducing the amount of effective information that can be extracted and thereby increasing the difficulty of fault classification and localization. When the fault resistance falls within the high-resistance regime of (1000, 2000] Ω, the accuracy drops more significantly.

G. Evaluation Under Intermittent Fault Conditions

Intermittent faults exhibit pronounced temporal variability and are more difficult to identify than conventional permanent faults. Therefore, this section further evaluates the proposed model under intermittent-fault conditions.

For intermittent faults, a representative intermittent-fault scenario is constructed in the IEEE 33-bus system to simulate the repeated ignition and extinction of arc faults under realistic operating conditions. The fault occurs periodically, persists only for a short duration, and then quickly returns to normal operation. This process is repeatedly triggered over multiple consecutive occurrences to reflect the typical repetitive nature of intermittent faults. Except for this fault configuration, all other experimental settings remain unchanged, and the corresponding sample set is generated accordingly. The intermittent-fault sample set is then used to evaluate each model, and Table VII presents the test results under intermittent-fault conditions.

Overall, intermittent faults significantly increase the difficulty of fault recognition, and all baseline models exhibit lower accuracy than in conventional fault scenarios. However, the proposed RGCN-NF achieves the best overall performance. Notably, the performance degradation of STGCN is relatively small, mainly due to its spatiotemporal convolution module, which effectively extracts temporal features and captures the short-term dynamic variations of signals during the ignition and extinction processes of arc faults.

TABLE VII
TEST-SET PERFORMANCE OF DIFFERENT MODELS UNDER INTERMITTENT FAULT CONDITIONS

Models	FCA	FLA	JA
RGCN-NF	91.23	**89.60**	**86.65**
STGCN [26]	**92.15**	86.62	83.04
GCN [23]	85.89	78.93	71.40
Fa-Mb-ResNet [15]	87.62	76.70	70.09
CNN [18]	80.82	71.16	62.18

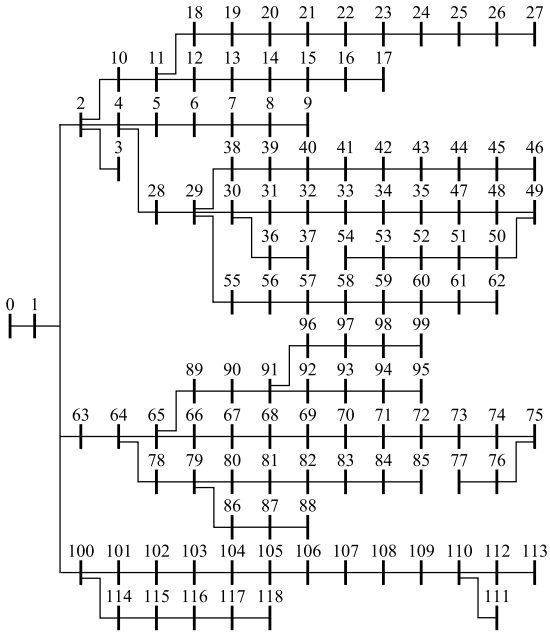


Fig. 12. Topology of the 119-bus test system.

H. Application to a Large-Scale Network

To further evaluate the applicability and scalability of the proposed framework on larger and more complex DNs, additional experiments are conducted on the 119-bus test system [46]. The topology of the 119-bus test system is illustrated in Fig. 12. To accommodate the larger network scale and longer topological paths, the receptive-field size k in EFP for degree-1 unmeasured nodes is set to 6. Apart from this adjustment, all other parameter settings of EFP and RGCN-NF remain identical to those used for the IEEE 33-bus system.

First, after data reconstruction using the EFP algorithm, the performance of RGCN-NF is compared with that of the baseline models on fault classification, fault localization, and the joint task, as summarized in Table VIII. The results show that, even with a substantially larger network, RGCN-NF maintains high accuracy across all tasks. Its overall performance remains close to that achieved on the IEEE 33-bus system, further validating the effectiveness and generalization capability of the neighbor-filtering mechanism and the node-level residual structure in mitigating oversmoothing and enabling deep representation learning.

Authorized licensed use limited to: XIDIAN UNIVERSITY. Downloaded on October 23,2026 at 05:39:18 UTC from IEEE Xplore. Restrictions apply.

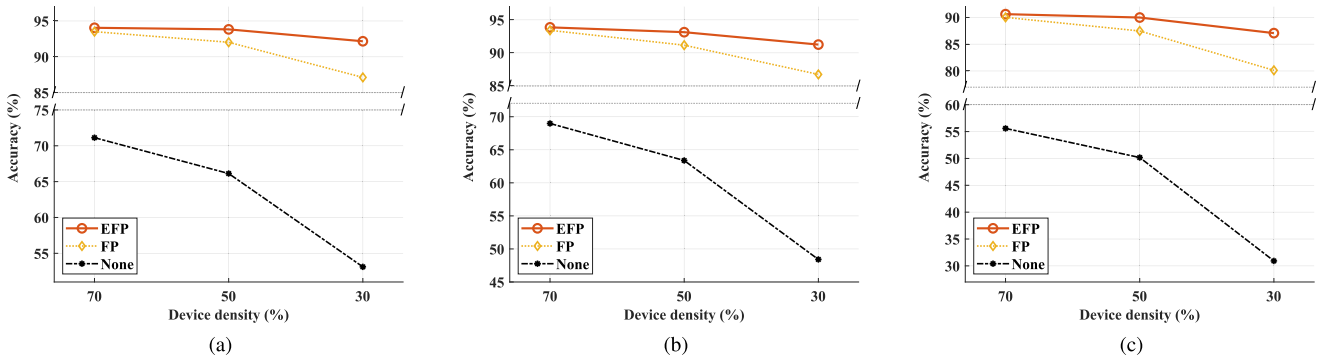


Fig. 13. Performance comparison of data completion methods under different measurement device densities on the 119-bus test system. (a) FCA. (b) FLA. (c) JA.

TABLE VIII
TEST-SET ACCURACY AND INFERENCE TIME OF DIFFERENT MODELS ON THE 119-BUS TEST SYSTEM

Models	FCA	FLA	JA	Time (s)
RGCN-NF	**93.79**	**93.02**	**89.31**	5.35×10^{-4}
STGCN [26]	89.10	80.77	75.23	8.13×10^{-4}
GCN [23]	85.62	80.36	73.39	$\mathbf{3.72\times10^{-5}}$
Fa-Mb-ResNet [15]	88.41	82.08	78.09	2.20×10^{-4}
CNN [18]	80.43	73.11	65.98	5.62×10^{-3}

TABLE IX
MODEL PERFORMANCE UNDER THE RING-OPERATED TOPOLOGY ON THE IEEE 33-BUS SYSTEM

Models	FCA	FLA	JA
RGCN-NF	**94.57**	**93.02**	**90.95**
STGCN [26]	92.04	85.91	83.11
GCN [23]	87.73	80.83	75.14
Fa-Mb-ResNet [15]	83.46	76.93	69.64
CNN [18]	80.60	70.59	61.78

Second, the effectiveness of the EFP algorithm in completing missing data in this large-scale network is evaluated. Based on the FCA, FLA, and JA achieved by RGCN-NF, three settings are compared: EFP, FP, and without data completion. As shown in Fig. 13, applying EFP significantly improves the performance of RGCN-NF compared with using incomplete data directly or employing the FP algorithm. These results indicate that the features reconstructed by EFP maintain strong generalization capability even in large and complex DNs.

I. Evaluation Under Ring-Operated Topology

To further verify the applicability and robustness of the proposed framework beyond radial topologies, a ring-operated scenario is constructed on the IEEE 33-bus system. Specifically, by closing the normally open tie-line switches, the standard radial topology is converted into a looped operating topology, under which the fault classification and fault localization performance of the proposed model is evaluated.

The IEEE 33-bus system includes five normally open tie lines connecting the bus pairs (7–20), (8–14), (11–21), (17–32), and (24–28). Based on the standard radial IEEE 33-bus topology, the switch status of these five tie lines is changed from normally open to normally closed, thereby forming a ring-operated topology. Except for the topology change and the resulting expansion of the candidate fault-line set, all other experimental settings are kept identical to those in the radial scenario.

Under the ring-topology scenario, the fault classification and fault localization performance of the proposed RGCN-NF and the baseline models are evaluated, and the results are summarized in Table IX. It can be observed that after introducing the ring topology, the proposed model still maintains high diagnostic performance and outperforms all baseline models in FCA, FLA, and JA. Further inspection shows that the performance of graph-based models, such as RGCN-NF, STGCN, and GCN, does not exhibit noticeable fluctuations compared with the radial DN, whereas nongraph models, such as Fa-Mb-ResNet and CNN, suffer a certain degradation in accuracy. This is mainly because the introduction of tie lines in the ring topology makes the network interconnections more complex. Graph models can explicitly exploit the topology and update adjacency relations accordingly for information aggregation, and are therefore more robust to topology changes. In contrast, nongraph models cannot explicitly capture the structural changes brought by the added tie lines, making their feature extraction and decision process more susceptible to topology variations.

V. CONCLUSION

To improve the accuracy of fault classification and localization in DNs under sparse measurements, we propose a novel two-stage framework that integrates the EFP algorithm with the RGCN-NF. First, the EFP algorithm assigns different receptive fields to different node types and constructs iterative equations based on the graph structure to reconstruct the data of unmeasured nodes due to sparse device deployment, thereby enriching local neighborhood information. In the second stage, the RGCN-NF enables deeper network architectures

Authorized licensed use limited to: XIDIAN UNIVERSITY. Downloaded on October 23,2026 at 05:39:18 UTC from IEEE Xplore. Restrictions apply.

without suffering from oversmoothing by selectively reweighting neighbor contributions and introducing node-level residual connections.

Comprehensive experiments on the IEEE 33-bus and 119-bus test systems demonstrate that, under various operating scenarios considering practical measurement uncertainties, the proposed framework achieves improvements in both fault classification and fault localization performance. It maintains excellent robustness even when measurements are unavailable at up to 70% of the nodes. In addition, the framework exhibits good generalization capability under randomly deployed measurement devices. These results confirm that combining the EFP algorithm with RGCN-NF is an effective strategy for improving fault classification and localization performance in DNs.

In future work, we will explore incorporating physics-constrained learning strategies inspired by impedance-based methods to further improve precise fault location. In addition, we will conduct a more in-depth investigation of composite faults and systematically evaluate, when current-channel measurements are available, the impact of incorporating current features on fault classification and localization performance.

REFERENCES

[1] M.-F. Guo, H. Lin, J. Lin, and Q. Hong, "Synthetic zero-sequence signal-based intelligent location method for single-phase ground fault in distribution networks," *IEEE Trans. Instrum. Meas.*, vol. 74, pp. 1–13, 2025.

[2] X. Lv et al., "An improved Bayesian learning method for distribution grid fault location and meter deployment optimization," *IEEE Trans. Instrum. Meas.*, vol. 73, pp. 1–10, 2024.

[3] F. Xiao, M. Wu, K. Song, T. Lu, and Q. Ai, "Diagnosis of distribution network fault using multiresolution S-transform and modified convolution neural network," *Int. J. Electr. Power Energy Syst.*, vol. 162, Nov. 2024, Art. no. 110294.

[4] X. Hu, J. Ma, R. Zhang, D. Ma, and Q. Wang, "Ts-GSAN: A two-stage graphical spatiotemporal attention network fault localization method for distributed energy systems," *IEEE Trans. Instrum. Meas.*, vol. 74, pp. 1–8, 2025.

[5] L. Tang et al., "Resilience enhancement of active distribution networks under extreme disaster scenarios: A comprehensive overview of fault location strategies," *Renew. Sustain. Energy Rev.*, vol. 189, Jan. 2024, Art. no. 113898.

[6] G. Luo, B. Shang, X. Wang, Z. Liu, C. Liu, and J. He, "Intelligent location method with limited measurement information for multibranch distribution networks," *IEEE Trans. Instrum. Meas.*, vol. 73, pp. 1–15, 2024.

[7] P. A. Pegoraro, C. Sitzia, A. V. Solinas, S. Sulis, D. Carta, and A. Benigni, "Improved fault detection and location method in three-phase distribution networks leveraging traceable PMU measurements," *IEEE Trans. Instrum. Meas.*, vol. 74, pp. 1–12, 2025.

[8] S. F. Alwash, V. K. Ramachandaramurthy, and N. Mithulananthan, "Fault-location scheme for power distribution system with distributed generation," *IEEE Trans. Power Del.*, vol. 30, no. 3, pp. 1187–1195, Jun. 2015.

[9] H. Mirshekali, R. Dashti, A. Keshavarz, A. J. Torabi, and H. R. Shaker, "A novel fault location methodology for smart distribution networks," *IEEE Trans. Smart Grid*, vol. 12, no. 2, pp. 1277–1288, Mar. 2021.

[10] F. Conte, F. D'Agostino, B. Gabriele, G.-P. Schiapparelli, and F. Silvestro, "Fault detection and localization in active distribution networks using optimally placed phasor measurements units," *IEEE Trans. Power Syst.*, vol. 38, no. 1, pp. 714–727, Jan. 2023.

[11] F.-X. Chen, M.-F. Guo, J. Lin, Y.-L. Zheng, X.-K. Zeng, and Q. Hong, "Single-ended traveling wave location for single-phase ground faults in distribution networks based on Hough transform," *IEEE Trans. Instrum. Meas.*, vol. 74, pp. 1–16, 2025.

[12] Y. Wang et al., "Fault location method for distribution network considering distortion of traveling wavefronts," *Int. J. Electr. Power Energy Syst.*, vol. 159, Aug. 2024, Art. no. 110065.

[13] K. Zhu and P. W. T. Pong, "Fault classification of power distribution cables by detecting decaying DC components with magnetic sensing," *IEEE Trans. Instrum. Meas.*, vol. 69, no. 5, pp. 2016–2027, May 2020.

[14] K. Dubey and P. Jena, "Novel fault detection & classification index for active distribution network using differential components," *IEEE Trans. Ind. Appl.*, vol. 60, no. 3, pp. 4530–4540, May 2024.

[15] L. Yang, Y. Li, and Z. Wei, "Fa-mb-ResNet for grounding fault identification and line selection in the distribution networks," *IEEE Internet Things J.*, vol. 9, no. 13, pp. 11115–11125, Jul. 2022.

[16] M.-F. Guo, N.-C. Yang, and W.-F. Chen, "Deep-learning-based fault classification using Hilbert–Huang transform and convolutional neural network in power distribution systems," *IEEE Sensors J.*, vol. 19, no. 16, pp. 6905–6913, Aug. 2019.

[17] M.-F. Guo, J.-H. Gao, X. Shao, and D.-Y. Chen, "Location of single-line-to-ground fault using 1-D convolutional neural network and waveform concatenation in resonant grounding distribution systems," *IEEE Trans. Instrum. Meas.*, vol. 70, pp. 1–9, 2021.

[18] M. N. I. Siddique, M. Shafiullah, S. Mekhilef, H. R. Pota, and M. A. Abido, "Fault classification and location of a PMU-equipped active distribution network using deep convolution neural network (CNN)," *Electric Power Syst. Res.*, vol. 229, Apr. 2024, Art. no. 110178.

[19] A. Ghaemi, A. Safari, H. Afsharirad, and H. Shayeghi, "Accuracy enhance of fault classification and location in a smart distribution network based on stacked ensemble learning," *Electric Power Syst. Res.*, vol. 205, Apr. 2022, Art. no. 107766.

[20] C. Li et al., "Takagi–sugeno fuzzy based power system fault section diagnosis models via genetic learning adaptive GSK algorithm," *Knowledge-Based Syst.*, vol. 255, Nov. 2022, Art. no. 109773.

[21] Q. Li et al., "Incipient fault detection in power distribution system: A time–frequency embedded deep-learning-based approach," *IEEE Trans. Instrum. Meas.*, vol. 72, pp. 1–14, 2023.

[22] W. Liao, B. Bak-Jensen, J. R. Pillai, Y. Wang, and Y. Wang, "A review of graph neural networks and their applications in power systems," *J. Modern Power Syst. Clean Energy*, vol. 10, no. 2, pp. 345–360, Mar. 2022.

[23] K. Chen, J. Hu, Y. Zhang, Z. Yu, and J. He, "Fault location in power distribution systems via deep graph convolutional networks," *IEEE J. Sel. Areas Commun.*, vol. 38, no. 1, pp. 119–131, Jan. 2020.

[24] J. T. de Freitas and F. G. F. Coelho, "Fault localization method for power distribution systems based on gated graph neural networks," *Electr. Eng.*, vol. 103, no. 5, pp. 2259–2266, Oct. 2021.

[25] Q.-H. Ngo, B. L. H. Nguyen, J. Zhang, K. Schoder, H. Ginn, and T. Vu, "Deep graph neural network for fault detection and identification in distribution systems," *Electric Power Syst. Res.*, vol. 247, Oct. 2025, Art. no. 111721.

[26] J. Hu et al., "Fault location and classification for distribution systems based on deep graph learning methods," *J. Modern Power Syst. Clean Energy*, vol. 11, no. 1, pp. 35–51, 2023.

[27] D. Chanda and N. Y. Soltani, "A heterogeneous graph-based multi-task learning for fault event diagnosis in smart grid," *IEEE Trans. Power Syst.*, vol. 40, no. 2, pp. 1427–1438, Aug. 2024.

[28] R. F. Buzo, H. M. Barradas, and F. B. Leão, "A new method for fault location in distribution networks based on voltage sag measurements," *IEEE Trans. Power Del.*, vol. 36, no. 2, pp. 651–662, Apr. 2021.

[29] L. Guo et al., "Topology and parameter identification of three-phase low-voltage distribution network and voltage control strategy based on user smart meter measurements: A data-driven approach," *IEEE Trans. Instrum. Meas.*, vol. 74, pp. 1–13, 2025.

[30] Q. Li, Z. Han, and X.-M. Wu, "Deeper insights into graph convolutional networks for semi-supervised learning," in *Proc. AAAI Conf. Artif. Intell.*, 2018, pp. 3538–3545.

[31] D. Chen, Y. Lin, W. Li, P. Li, J. Zhou, and X. Sun, "Measuring and relieving the over-smoothing problem for graph neural networks from the topological view," in *Proc. AAAI Conf. Artif. Intell.*, Apr. 2020, pp. 3438–3445.

[32] T. Konstantin Rusch, M. M. Bronstein, and S. Mishra, "A survey on oversmoothing in graph neural networks," 2023, *arXiv:2303.10993*.

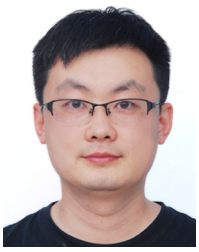
[33] S. S. Gururajapathy, H. Mokhlis, and H. A. Illias, "Fault location and detection techniques in power distribution systems with distributed generation: A review," *Renew. Sustain. Energy Rev.*, vol. 74, pp. 949–958, Jul. 2017.

[34] E. Rossi, H. Kenlay, M. I. Gorinova, B. P. Chamberlain, X. Dong, and M. M. Bronstein, "On the unreasonable effectiveness of feature propagation in learning on graphs with missing node features," in *Proc. Learn. graphs Conf.*, 2021, pp. 1–11.

[35] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," 2016, *arXiv:1609.02907*.

Authorized licensed use limited to: XIDIAN UNIVERSITY. Downloaded on October 23,2026 at 05:39:18 UTC from IEEE Xplore. Restrictions apply.

[36] M. E. Baran and F. F. Wu, "Network reconfiguration in distribution systems for loss reduction and load balancing," *IEEE Trans. Power Del.*, vol. 4, no. 2, pp. 1401–1407, Apr. 1989.
[37] P. A. Pegoraro, C. Sitzia, A. V. Solinas, S. Sulis, D. Carta, and A. Benigni, "Compensation of systematic errors for improved PMU-based fault detection and location in three-phase distribution grids," *IEEE Trans. Instrum. Meas.*, vol. 73, pp. 1–12, 2024.
[38] P. A. Pegoraro, K. Brady, P. Castello, C. Muscas, and A. von Meier, "Compensation of systematic measurement errors in a PMU-based monitoring system for electric distribution grids," *IEEE Trans. Instrum. Meas.*, vol. 68, no. 10, pp. 3871–3882, Oct. 2019.
[39] P. A. Pegoraro, C. Sitzia, A. V. Solinas, and S. Sulis, "Transmission line parameters estimation in the presence of realistic PMU measurement error models," *Meas.*, vol. 218, Aug. 2023, Art. no. 113175.
[40] *IEEE/IEC International Standard—Measuring Relays and Protection Equipment—Part 118-1: Synchrophasor for Power Systems—Measurements*, Standard 60255-118-1, 2018, pp. 1–78.
[41] K. You, M. Long, J. Wang, and M. I. Jordan, "How does learning rate decay help modern neural networks?," 2019, *arXiv:1908.01878*.
[42] N. Shirish Keskar, D. Mudigere, J. Nocedal, M. Smelyanskiy, and P. Tak Peter Tang, "On large-batch training for deep learning: Generalization gap and sharp minima," 2016, *arXiv:1609.04836*.
[43] G. E. Hinton, N. Srivastava, A. Krizhevsky, I. Sutskever, and R. R. Salakhutdinov, "Improving neural networks by preventing co-adaptation of feature detectors," 2012, *arXiv:1207.0580*.
[44] X. Wang, X. Wang, W. Liu, J. Gao, and X. Wei, "Faulty feeder detection under high-impedance fault for active distribution networks in resonant grounding mode," *IEEE Trans. Instrum. Meas.*, vol. 73, pp. 1–11, 2024.
[45] K. Liao, Y. Tan, B. Li, and Z. He, "High impedance fault detection in resonant-grounded distribution networks using continuous wavelet transform and deep residual networks," *IEEE Trans. Instrum. Meas.*, vol. 74, pp. 1–16, 2025.
[46] D. Zhang, Z. Fu, and L. Zhang, "An improved TS algorithm for loss-minimum reconfiguration in large-scale distribution systems," *Electric Power Syst. Res.*, vol. 77, nos. 5–6, pp. 685–694, Apr. 2007.



Wei Sun (Senior Member, IEEE) received the B.E. degree in automation, the M.S. degree in detection technology and automatic equipment, and the Ph.D. degree in electrical engineering from Hefei University of Technology, Hefei, China, in 2004, 2007, and 2012, respectively.

He is currently a Professor with Hefei University of Technology. His research interests include wireless sensor networks, networked control systems, and smart grids.



Song Liu received the B.S. degree in automation from Changsha University of Science and Technology, Changsha, China, in 2023. He is currently pursuing the M.S. degree in control engineering with Hefei University of Technology, Hefei, China.

His research interests include synchrophasor measurement, fault diagnosis in distribution networks (DNs), and machine-learning applications in power systems.



Hengfeng Zhang received the B.S. degree in automation from Anhui Jianzhu University, Hefei, China, in 2022. He is currently pursuing the Ph.D. degree with the School of Electrical Engineering and Automation, Hefei University of Technology, Hefei.

His current research interests include power system data analytics, topology identification, and fault diagnosis.



Zhiqiang Zhang received the B.S. degree in automation from Hefei University of Technology, Hefei, China, in 2016, and the Ph.D. degree in control science and control engineering from the Department of Automation, Xi'an Jiaotong University, Xi'an, China, in 2022.

Since 2022, he has been a Master Supervisor with the School of Electrical Engineering and Automation, Hefei University of Technology. His research interests include fault diagnosis, intelligent operation and maintenance, and new power systems.



Fei Lin received the M.S. degree from Hefei University of Technology, Hefei, China, in 2006.

She is currently with the Energy Measurement Department, Marketing Service Center, State Grid Anhui Electric Power Company Ltd., Hefei. Her main research interests include electrical measurement technology and management.

Authorized licensed use limited to: XIDIAN UNIVERSITY. Downloaded on October 23,2026 at 05:39:18 UTC from IEEE Xplore. Restrictions apply.